



E22-400M30S Product Datasheet

SX1268 433/470MHz 1W SPI SMD LoRa Module



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Chapter 1 Overview

1.1 Introduction

The E22-400M30S is a self-developed LoRa™ wireless module based on the new-generation LoRa™ RF chip SX1268 from Semtech USA. It boasts a maximum power of 1W and is suitable for 433/470MHz surface-mount LoRa™ wireless communications. It utilizes an industrial-grade, high-precision 32MHz crystal oscillator.

Utilizing the imported SX1268 as its core, it integrates a built-in power amplifier (PA) and low-noise amplifier (LNA), achieving a maximum transmit power of 1W while further enhancing receive sensitivity. This significantly improves overall communication stability compared to products without PAs and LNAs. Compared to previous-generation LoRa™ transceivers, it boasts enhanced anti-interference performance and communication range, further widening the gap with products using FSK and GFSK modulation schemes.

This product covers an ultra-wide applicable frequency range of 410-493MHz and is backward compatible with the SX1278 and SX1276. Since this module is a pure RF transceiver module, it needs to be driven by an MCU or a dedicated SPI debugging tool.



1.2 Features

- Compared to the SX1278 module, the SX1268 module offers significant advantages: lower power consumption, faster speed, and longer range.
- Under ideal conditions, the communication range can reach 12 km.
- A built-in PA + LNA significantly improves communication range and stability.
- A maximum transmit power of 1W, software-adjustable in multiple levels.
- Supports the global license-free ISM 433/470MHz frequency bands.
- In LoRa™ mode, it supports data rates from 0.018kbps to 62.5kbps.
- In FSK mode, it supports data rates up to 300kbps.
- Backwards compatible with the SX1278/SX1276 series RF transceivers.
- Large FIFO capacity, supporting 256-byte data buffering.
- Introducing the new SF5 spreading factor for dense network support.
- Supports 2.5-5.5V power supply, and optimal performance is guaranteed with power supplies greater than 5V.
- Industrial-grade standard design supports extended use in temperatures from -40°C to +85°C.
- Optional dual antennas (IPEX/stamp hole) facilitate user development and integration.

1.3 Application Scenarios

- Home security alarms and remote keyless entry;
- Smart home and industrial sensors;
- Wireless alarm and security systems;
- Building automation solutions;
- Wireless industrial remote controls;
- Advanced Metering Infrastructure (AMI);
- Automotive applications.

Chapter 2 Specifications

2.1 Absolute maximum ratings

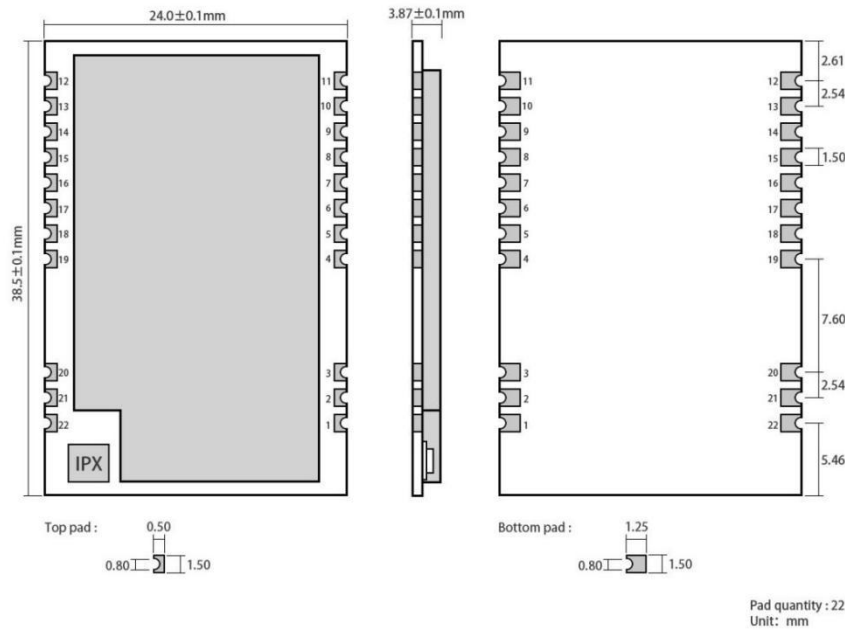
Main parameters	Value		Note
	Minimum	Maximum	
Supply Voltage (V)	0	5.5	Over 5.5 V will permanently damage the module.
Blocking Power (dBm)	-	10	Probability of burnout is low when used at close range.
Operating Temperature (°C)	- 40	+85	Industrial Grade

2.2 Operating parameters

Parameter	Value			Note
	Minimum	Typical	Maximum	
Operating Voltage (V)	2.5	5.0	5.5	≥5.0V can guarantee output power
Communication Level (V)		3.3		When using 5V TTL, it is recommended to add level conversion
Operating Temperature (°C)	- 40	-	+85	Industrial-grade design
Operating Frequency Band (MHz)	410	433/470	493	Supports ISM bands.
Power consumption	Transmit Current (mA)	650		Instantaneous power consumption
	Receive Current (mA)	14		
	Sleep Current (uA)	2		Software shutdown
Maximum Transmit Power (dBm)	29.5	30.0	31	
Receive Sensitivity (dBm)	- 149	- 150	- 151	At 0.3kbps air data rate
Air data rate (bps)	0.6k	-	300k	User programmable control
	0.018k	-	62.5k	User programmable control

Parameter	Value	Note
Reference distance	12000m	Clear and open air, antenna gain: 5 dBi, antenna height: 2.5 meters, air data rate: 0.3 kbps
FIFO	256Bytes	Maximum single transmission length
Crystal oscillator frequency	32MHz	
Modulation	LoRa (recommended)	
Packaging	SMD	
Interface	2.54mm	Postage stamp holes
Communication interface	SPI	Data rate 0-10 Mbps
Dimensions	38.5*24mm	
Antenna interface	Postage stamp hole/IPEX	Equivalent impedance: approximately 50 Ω

Chapter 3 Mechanical Dimensions and Pin Definitions



Pin number	Pin Name	Direction type	Description
1	GND		Ground wire, connected to the power supply reference ground.
2	GND		Ground wire, connected to the power supply reference ground.
3	GND		Ground wire, connected to the power supply reference ground.
4	GND		Ground wire, connected to the power supply reference ground.
5	GND		Ground wire, connected to the power supply reference ground.
6	RXEN	Input	The RF switch receive control pin is connected to an external MCU IO port. For logic control, refer to Table 1 below. For control examples, refer to the EBYTE official website demo.
7	TXEN	Input	The RF switch receive control pin is connected to an external MCU IO port. For logic control, refer to Table 1 below. For control examples, refer to the EBYTE official website demo.
8	DIO2	Input/Output	Configurable general-purpose IO port (see the SX1268 manual for details).
9	VCC	-	Power supply range: 2.5-5.5V (adding an external ceramic filter capacitor is recommended).
10	VCC	-	Power supply range: 2.5-5.5V (adding an external ceramic filter capacitor is recommended).
11	GND	-	Ground wire, connected to the power supply reference ground.
12	GND	-	Ground wire, connected to the power supply reference ground.
13	DIO1	Input/Output	Configurable general-purpose IO port (see the SX1268 manual for details).
14	BUSY	Output	Used for status indication (see the SX1268 manual for details).
15	NRST	Input	Chip reset trigger input pin, active low.
16	MISO	Output	SPI data output pin.
17	MOSI	Input	SPI data input pin.
18	SCK	Input	SPI clock input pin.
19	NSS	Input	Module chip select pin, used to start an SPI communication.
20	GND	-	Ground wire, connected to the power supply reference ground.
21	ANT	-	Antenna interface, stamp hole (50Ω characteristic impedance). Because the module has a built-in PA, there is a magnification relationship between the output power of the SX1268 RF chip (pre-stage) and the actual module output power. See Table 2 below.
22	GND	-	Ground wire, connected to the power supply reference ground.
-	DIO3	Input/Output	Used internally in the module to power the 32MHZ TCXO crystal oscillator (DIO3 must be configured to output 2.2V)

Chapter 4 Basic Operations

4.1 Hardware Design

- It is recommended to power this module using a DC regulated power supply with minimal ripple. The module must be reliably grounded.
- Ensure the positive and negative poles of the power supply are connected correctly. Reverse connection may cause permanent damage to the module.
- Check the power supply to ensure it is within the recommended voltage range. Exceeding the maximum voltage may cause permanent damage to the module.
- Check the power supply stability to ensure the voltage does not fluctuate significantly.
- When designing the power supply circuit for the module, it is generally recommended to leave at least 30% margin to ensure long-term stable operation of the entire device.
- The module should be placed as far away as possible from areas with high electromagnetic interference, such as power supplies, transformers, and high-frequency traces.
- High-frequency digital traces, high-frequency analog traces, and power traces must be routed away from underneath the module. If absolutely necessary, route them underneath the module. Assuming the module is soldered to the top layer, lay ground copper on the top layer of the module contact area (cover all areas with copper and ensure a good ground connection). Route the traces close to the digital portion of the module on the bottom layer.
- Assuming the module is soldered or placed on the top layer, lay ground copper on the bottom layer of the module contact area. Random routing on the 10-layer or other layers is also incorrect and will affect the module's spurious emissions and receiver sensitivity to varying degrees.
- If there are devices with significant electromagnetic interference around the module, this will significantly impact module performance. Depending on the intensity of the interference, it is recommended to keep them away from the module. If possible, appropriate isolation and shielding can be implemented.
- If there are traces with significant electromagnetic interference around the module (high-frequency digital, high-frequency analog, and power traces), this will significantly impact module performance. Depending on the intensity of the interference, it is recommended to keep them away from the module. If possible, appropriate isolation and shielding can be implemented.
- If the communication line uses a 5V voltage level, a 1k-5.1k resistor must be connected in series (not recommended, as there is a risk of damage).
- Keep them away from some 2.4GHz TTL protocols, such as USB 3.0.
- The antenna mounting structure significantly affects module performance. Ensure the antenna is exposed, preferably pointing vertically. When the module is installed inside a housing, use a high-quality antenna extension cable to extend the antenna outside the housing.
- Never install the antenna inside a metal enclosure, as this will significantly reduce the transmission distance.
- It is recommended to add a 200Ω protective resistor to the RXD/TXD terminals of the external MCU.

4.2 Software development

- This module is an SX1268/SX1262 + PA + LNA. Its driver mode is identical to the SX1268/SX1262. Users can operate it according to the SX1268/SX1262 chip manual.
- DIO1 and DIO2 are general-purpose I/O ports that can be configured for various functions. DIO2 can be connected to TXEN and is not connected to an MCU I/O port. It is used to control the RF switch transmission. For details, see the SX1262 manual. If not used, it can be left floating.
- DIO3 is used internally to power the 32MHz TCXO crystal oscillator (DIO3 is configured to output 2.2V).

Table 1 [Transceiver state control truth table]

Radio State Pin	RXEN (Input)	TXEN (Input)	Remark
Transmit	0	1	"0": low level "1": high level
Receive	1	0	* RXEN, TXEN cannot be at the same time as "1".

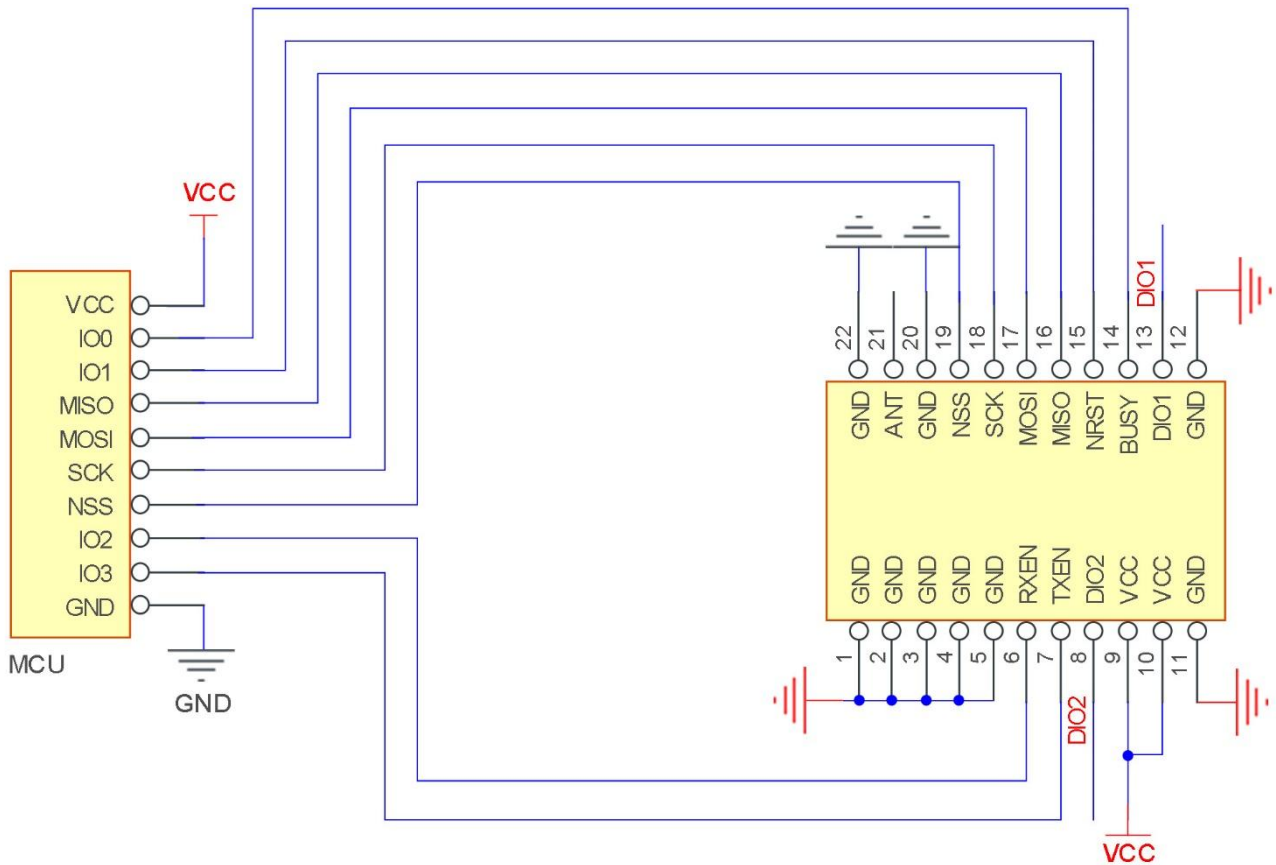
Low power consumption	0	0	* When switching to the transmit state, please delay at least 2ms before transmitting the data.
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Table 2 [The RF output power control table]

Set the power dBm (SX1268 Front-stage output)	Real power dBm (Module actual output)	Remark
18	29.96	* Test frequency was 433 MHz. * There are deviation in the actual power of each module, and this table value is used as a reference only. * The set power of this table will only take some typical values.
14	27.25	
10	24.02	
7	21.11	
4	18.08	
1	15.02	
- 2	12.29	
- 5	9.77	

Chapter 5 Basic Applications

5.1 Basic Circuit



Chapter 6 Frequently Asked Questions

6.1 Transmission distance is not ideal

- When there are direct line obstacles, the communication distance will be reduced accordingly.
- Temperature, humidity, and co-channel interference can increase the packet loss rate.
- The ground absorbs and reflects radio waves, so testing near the ground is poor.
- Seawater has a strong ability to absorb radio waves, so testing near the seashore is poor.
- Metal objects near the antenna, or the antenna is placed in a metal casing, will cause severe signal attenuation.
- The power register is incorrectly set, or the air rate is set too high (the higher the air rate, the closer the distance).
- The power supply voltage is below the recommended value at room temperature; lower voltage means lower power.
- The antenna is poorly matched to the module or the antenna itself has quality issues.

6.2 Modules are easily damaged

- Please check the power supply to ensure it is within the recommended voltage range. Exceeding the maximum voltage may cause permanent damage to the module.
- Please check the power supply stability to ensure the voltage does not fluctuate significantly.
- Please ensure anti-static operation during installation and use, as high-frequency components are sensitive to static electricity.
- Please ensure that the humidity is not too high during installation and use, as some components are humidity-sensitive.
- Unless otherwise specified, it is not recommended to use the module in extremely high or low temperatures.

6.3 Bit Error Rate Too High

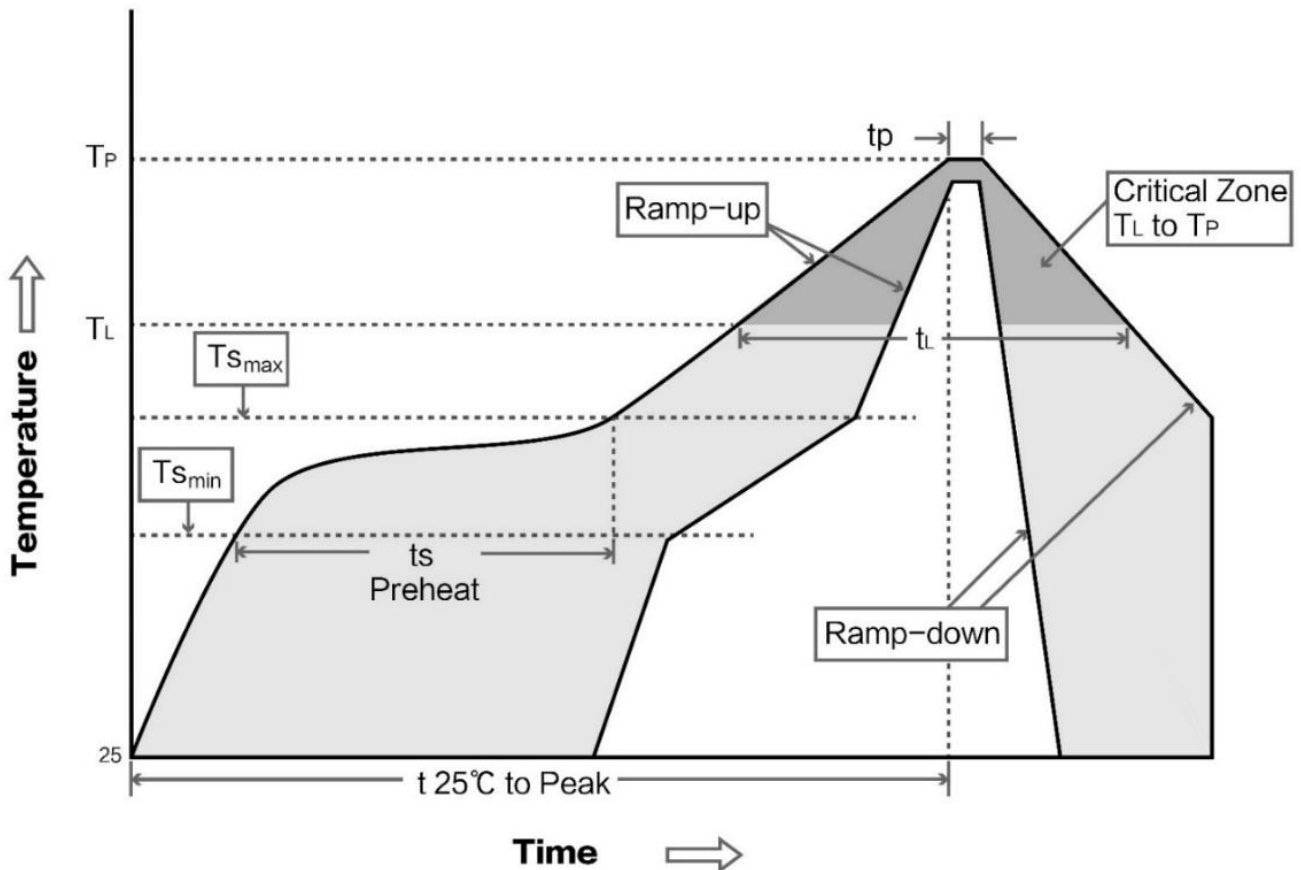
- There is co-frequency signal interference nearby. Move away from the interference source or change the frequency or channel to avoid interference.
- The SPI clock waveform is non-standard. Check for interference on the SPI line. The SPI bus trace should not be too long.
- A poor power supply may also cause garbled data. Ensure the reliability of the power supply.
- Poor-quality or overly long extension cables or feeders can also cause a high bit error rate.

Chapter 7 Soldering Instructions

7.1 Reflow Temperature

Profile Feature	Curve characteristics	Sn-Pb Assembly	Pb-Free Assembly
Solder Paste	Solder Paste	Sn63/Pb37	Sn96.5/Ag3/Cu0.5
Preheat Temperature (min) (T _{smin})	Minimum preheat temperature	100°C	150°C
Preheat Temperature (max) (T _{smax})	Maximum preheat temperature	150°C	200°C
Preheat Time (T _{smin} to T _{smax}) (ts)	Preheat time	60- 120 sec	60- 120 sec
Average ramp-up rate (T _{smax} to T _p)	Average ramp-up rate	3°C/second max	3°C/second max
Liquidous Temperature (T _L)	Liquidus temperature	183°C	217°C
Time (t _L) Maintained Above (T _L)	Time above liquidus	60- 90 sec	30- 90 sec
Peak temperature (T _p)	Peak temperature	220- 235°C	230- 250°C
Average ramp-down rate (T _p to T _{smax})	Average ramp-down rate	6°C/second max	6°C/second max
Time 25°C to peak temperature	Time from 25°C to peak temperature	6 minutes max	8 minutes max

7.2 Reflow Oven Curve



Chapter 8 Related Models

Product Model	Chip Solution	Frequency Hz	Transmit power, dBm	Distance km	Package	Size mm	Communication interface
E22-400M22S	SX1268	433/470M	22	7	SMD	14*20	SPI
E22-900M22S	SX1262	868/915M	22	7	SMD	14*20	SPI
E22-400M30S	SX1268	433/470M	30	12	SMD	24*38.5	SPI
E22-900M30S	SX1262	868/915M	30	12	SMD	24*38.5	SPI
E22-230T22S	SX1262	230M	22	5	SMD	16*26	TTL
E22-400T22S	SX1268	433/470M	22	5	SMD	16*26	TTL
E22-900T22S	SX1262	868/915M	22	5	SMD	16*26	TTL
E22-230T30S	SX1262	230M	30	10	SMD	25*40.5	TTL
E22-400T30S	SX1268	433/470M	30	10	SMD	25*40.5	TTL
E22-900T30S	SX1262	868/915M	30	10	SMD	25*40.5	TTL

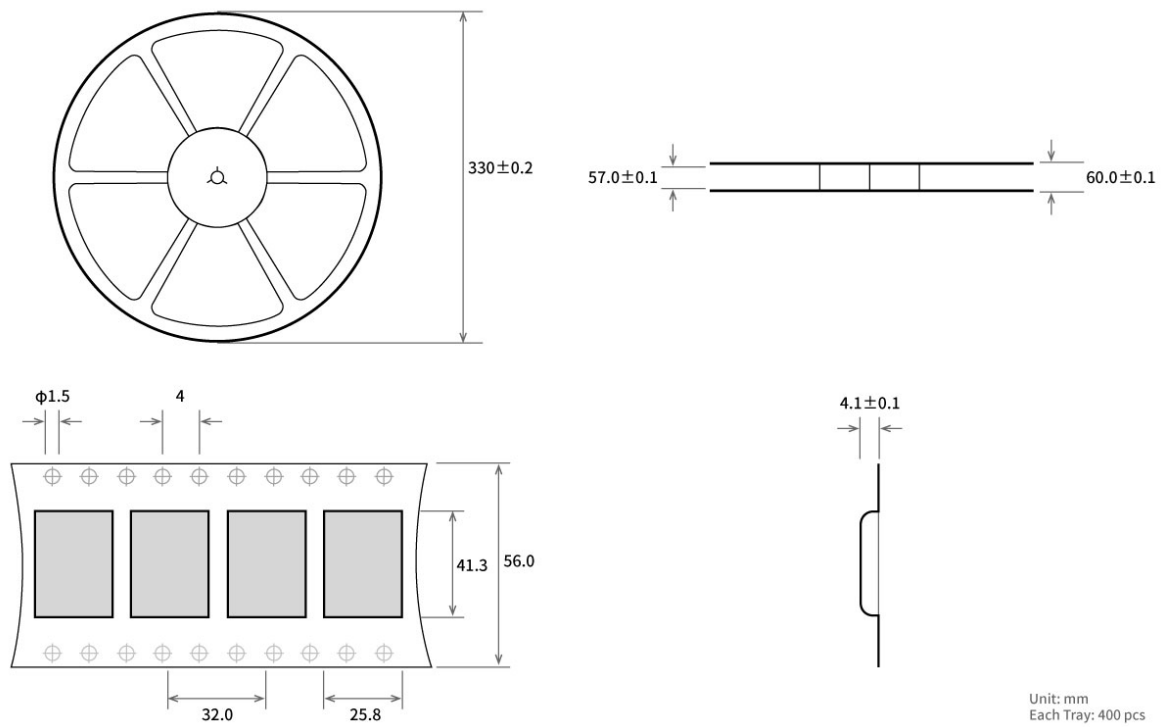
Chapter 9 Antenna Guide

9.1 Antenna Recommendations

Antennas play an important role in the communication process. Often, poor-quality antennas will have a great impact on the communication system. Therefore, our company recommends some antennas as supporting antennas for our wireless modules with excellent performance and reasonable prices.

Product Model	Type	Frequency band, Hz	Interface	Gain dBi	Dimensions mm	Feeder cm	Features
TX433-NP-4310	Flexible	433M	Soldering	2.0	43.8*9.5	-	Built-in flexible, FPC soft antenna
TX433-JZ-5	Rubber Stick	433M	SMA-J	2.0	52	-	Ultra-short straight, omnidirectional antenna
TX433-JZG-6	Rubber Stick	433M	SMA-J	2.5	62	-	Ultra-short straight, omnidirectional antenna
TX433-JW-5	Rubber Stick	433M	SMA-J	2.0	50	-	Bendable rubber stick, omnidirectional antenna
TX433-JWG-7	Rubber Stick	433M	SMA-J	2.5	75	-	Bendable rubber stick, omnidirectional antenna
TX433-JK-11	Rubber Stick	433M	SMA-J	2.5	110	-	Bendable rubber stick, omnidirectional antenna
TX433-JK-20	Rubber Stick	433M	SMA-J	3.0	210	-	Bendable rubber stick, omnidirectional antenna
TX433-XPL-100	Suction Cup	433M	SMA-J	3.5	185	100	Small suction cup antenna, cost-effective
TX433-XP-200	Suction Cup	433M	SMA-J	4.0	190	200	Neutral suction cup antenna, low loss
TX433-XPH-300	Suction Cup	433M	SMA-J	6.0	965	300	Large suction cup antenna, high gain
TX490-JZ-5	Rubber Stick	470/490M	SMA-J	2.0	50	-	Ultra-short straight, omnidirectional antenna
TX490-XPL-100	Suction Cup	470/490M	SMA-J	3.5	120	100	Small suction cup antenna, cost-effective

Chapter 10 Bulk Packaging Methods



Revision History

Version	Revision Date	Revision Notes	Maintainer
1.0	2018-09-18	Manual Released	huaa
1.1	2019-02-17	Bug Fixes	Ray
1.2	2019-03-22	Content Updates	Ray
1.3	2019-09-24	Content Updates	Ren
1.4	2020-04-14	Bug Fixes	Ren
1.5	2020-11-26	Bug Fixes	Linson
1.6	2024-11-20	Added description of RF transceiver logic control and front-end power output	Ning
1.7	2024-12-31	Deleted certification description	Lei
1.8	2025-3-3	Modified actual product images	LAU

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